

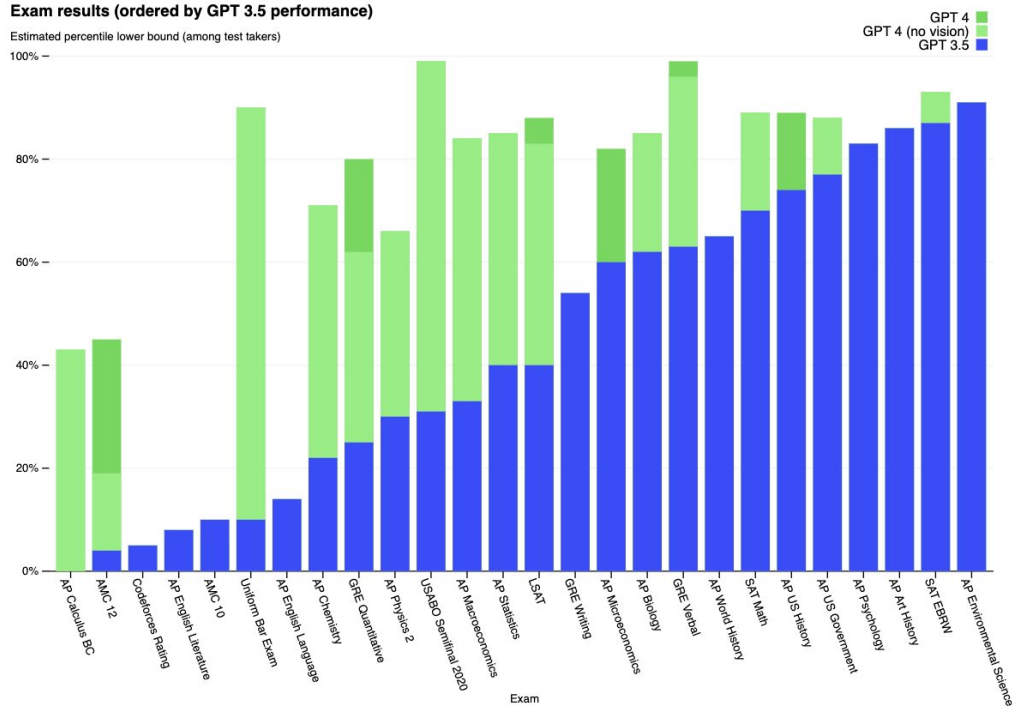


Figure 1: Taken directly from GPT-4 Technical Report (OpenAI, 2023b). To get a sense of how quickly model capabilities are progressing – consider the jump in exam performance between GPT-3.5 and GPT-4 (OpenAI, 2023b).

Our study is motivated less by the progress of these models alone though, and more by the breadth, scale, and capabilities we’ve seen in the complementary technologies developed around them. The role of complementary technologies remains to be seen, but maximizing the impact of LLMs appears contingent on integrating them with larger systems (Bresnahan, 2019; Agrawal et al., 2021). While the focus of our discussion is primarily on the generative capabilities of LLMs, it is important to note that these models can also be utilized for various tasks beyond text generation. For example, embeddings from LLMs can be used for custom search applications, and LLMs can perform tasks such as summarization and classification where the context may be largely contained in the prompt.

To complement predictions of technology’s impacts on work and provide a framework for understanding the evolving landscape of language models and their associated technologies, we propose a new rubric for assessing LLM capabilities and their potential effects on jobs. This rubric (A.1) measures the overall exposure of tasks to LLMs, following the spirit of prior work on quantifying exposure to machine learning (Brynjolfsson et al., 2018; Felten et al., 2018; Webb, 2020). We define exposure as a proxy for potential economic impact without distinguishing between labor-augmenting or labor-displacing effects. We employ human annotators and GPT-4 itself as a classifier to apply this rubric to occupational data in the U.S. economy, primarily sourced from the O*NET database.¹²

To construct our primary exposure dataset, we collected both human annotations and GPT-4 classifications, using a prompt tuned for agreement with a sample of labels from the authors. We observe similar agreement

¹This is distinct from recent social science research that makes use of LLMs to simulate human behavior (Horton, 2023; Sorensen et al., 2022)

²While our exposure rubric does not necessarily tie the concept of language models to any particular model, we were strongly motivated by our observed capabilities of GPT-4 and the suite of capabilities we saw in development with OpenAI’s launch partners (OpenAI, 2023b).

of LLMs on the workforce, policymakers and stakeholders can make more informed decisions to navigate the complex landscape of AI and its role in shaping the future of work.

7.1 LLM Conclusion (GPT-4's Version)

Generative Pre-trained Transformers (GPTs) generate profound transformations, garnering potential technological growth, permeating tasks, greatly impacting professions. This study probes GPTs' potential trajectories, presenting a groundbreaking rubric to gauge tasks' GPT exposure, particularly in the U.S. labor market.

7.2 LLM Conclusion (Author-Augmented Version)

Generative Pre-trained Transformers (GPTs) generate profound transformations, garnering potential technological growth, permeating tasks, gutting professional management. Gauging possible trajectories? Generate pioneering taxonomies, gather policymakers together, generalize past today.

Acknowledgments

Thank you to the group of annotators who helped us annotate task exposure, including Muhammad Ahmed Saeed, Bongane Zitha, Merve Özen Şenen, J.J., and Peter Hoeschele. We also thank Lauryn Fuld, Ashley Glat, Michael Lampe, and Julia Susser for excellent research assistance. We thank Miles Brundage for significant feedback on this paper.

We thank Todor Markov and Vik Goel for setting up the infrastructure we use to run our rubrics with GPT-4. We thank Lama Ahmad, Donald Bakong, Seth Benzell, Erik Brynjolfsson, Parfait Eloundou-Enyegue, Carl Frey, Sarah Giroux, Gillian Hadfield, Johannes Heidecke, Alan Hickey, Eric Horvitz, Shengli Hu, Ashyana Kachra, Christina Kim, Katya Klinova, Daniel Kokotajlo, Gretchen Krueger, Michael Lampe, Aalok Mehta, Larissa Schiavo, Daniel Selsam, Sarah Shoker, Prasanna Tambe, and Jeff Wu for feedback and edits at various stages of the project.

LLM assistance statement

GPT-4 and ChatGPT were used for writing, coding, and formatting assistance in this project.

A Rubric

A.1 Exposure

E Exposure Taxonomy

Consider the most powerful OpenAI large language model (LLM) This model can complete many tasks that can be formulated as having text input and text output where the context for the input can be captured in 2000 words. The model also cannot draw up-to-date facts (those from <1 year ago) unless they are captured in the input.

Assume you are a worker with an average level of expertise in your role trying to complete the given task. You have access to the LLM as well as any other existing software or computer hardware tools mentioned in the task. You also have access to any commonly available technical tools accessible via a laptop (e.g. a microphone, speakers, etc.). You do not have access to any other physical tools or materials.

Please label the given task according to the taxonomy below.

E0 – No exposure